# Solutions

## April 2018

Q1/ (a) Net premium $= 100{,}000 \dfrac{A_{45:\overline{20}|}}{\ddot{a}_{45:\overline{20}|}}$ ($A_{45:\overline{20}|} = 0.46998$, $\ddot{a}_{45:\overline{20}|} = 13.780$)

$= £3{,}410.60$ per year

$V(14) = 100{,}000\, A_{59:\overline{6}|} - 3410.60\, \ddot{a}_{59:\overline{6}|}$ ($A_{59:\overline{6}|} = 0.79446$, $\ddot{a}_{59:\overline{6}|} = 5.344$)

$= £61{,}219.78$

$V(13.7)\,[1.04]^{0.3} = {}_{0.3}q_{58.7} \times 100{,}000 + {}_{0.3}p_{58.7} \times 61{,}219.78$

$= £61{,}293.84$

$p_{58} = 0.993648$

${}_{0.3}p_{58.7} = (\,\text{''}\,)^{0.3} = 0.99809$

${}_{0.3}q_{58.7} = 1 - \text{''} = 0.00191$

$V(13.7) = £60{,}576.87$

(b) - Because there is a fixed valuation date for accounts (e.g. 31 December / March), but policies are sold during year.
- [or for a policy alteration / surrender etc]

Q2

(i) $CDR = \frac{75}{5,400} = 0.013889$

(ii) $CDR = \frac{1,385,000}{125,000,000} = 0.01108$

(iii) Expected deaths: $\frac{1600 \times 105\pi}{50m} = 3$

$+11$
$+69$
$83$

Ratio $= 75/83 = 0.898732$

(iv) The company has a low ratio, but the company has a higher CDR.

This appears to be because they are older than the population.

**Q3**

(a) Deferred period:

The period of time that a policyholder must remain sick before benefits commence.

This reduces administration, and reduces the possibly large volume of short-term claims.

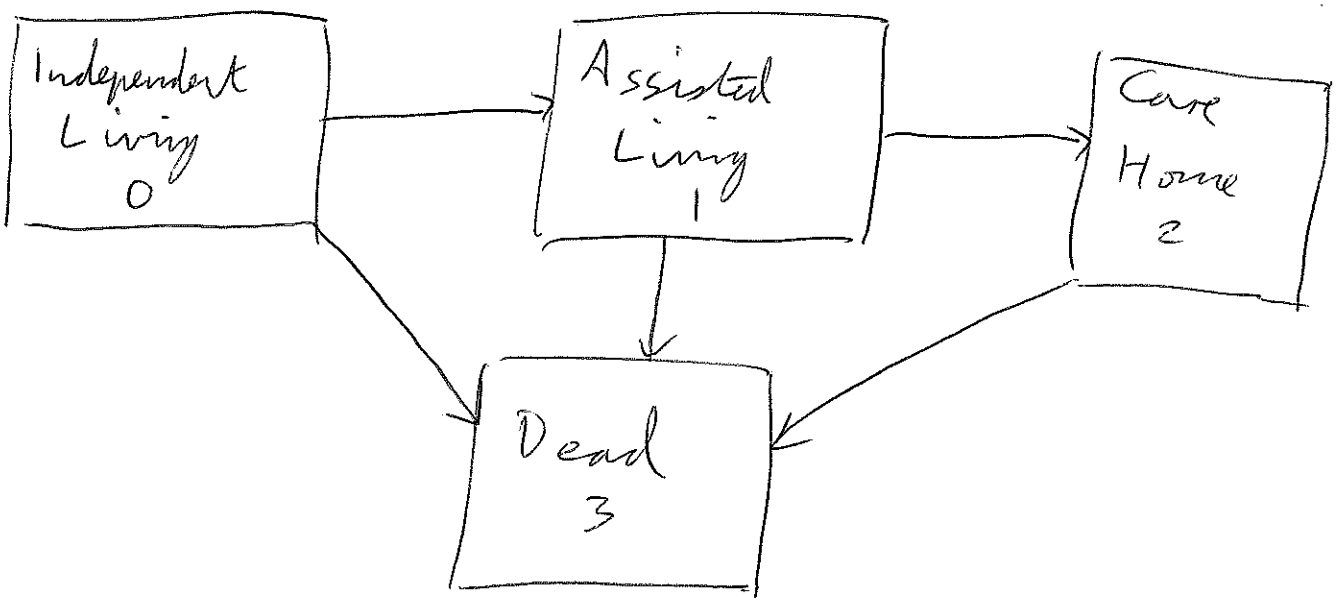
(b) Off-period:

Suppose that the benefits reduce after some period of continuous sickness.

Then the off-period defines the minimum period of good health that must pass before two episodes of sickness can be considered different.

This prevents policyholders from "recovering" for a short period, in order to regain the higher level of benefit.

Q4 (a)




(i) $${}_{10}p_x^{01} = \int_0^{10} {}_t p_x^{00} \, \mu_{x+t}^{01} \, {}_{10-t}p_x^{11} \, dt$$ (1)

Note:
$${}_t p_x^{00} = {}_t p_x^{\overline{00}}$$
$${}_t p_x^{11} = {}_t p_x^{\overline{11}}$$
here

Now $${}_t p_x^{00} = e^{-(\mu^{01} + \mu^{03})t}$$
$$= e^{-0.05t}$$

and $${}_{(10-t)}p_x^{11} = e^{-(\mu^{12} + \mu^{13})(10-t)}$$
$$= e^{-0.5(10-t)} = e^{-5} \cdot e^{0.5t}$$

$\therefore$ (1) becomes $${}_{10}p_x^{01} = \int_0^{10} e^{-0.05t} \cdot (0.02) \, e^{-5} \cdot e^{0.5t} \, dt$$

$$= 0.02 \cdot e^{-5} \int_0^{10} e^{0.45t} \, dt$$
$$= 0.02 \, e^{-5} \left[ e^{0.45t} / 0.45 \right]_0^{10}$$
$$= 0.02/0.45 \, e^{-5} (e^{4.5} - 1)$$
$$= \underline{\underline{0.002008}} = \underline{0.02666}$$

Q5

$$ {}_t p_x = \exp\left\{ -\int_0^t \mu_{x+s}\, ds \right\} $$

$$ = \exp\left\{ -\int_0^t k\, ds \right\} \quad \text{where } k = 0.04 \text{ or } 0.03 \text{ or } 0.07 \text{ for joint} $$

$$ = \exp(-kt) $$

(a)

$$ \bar{A}_{\overline{60:57}} = \bar{A}_{60} + \bar{A}_{57} - \bar{A}_{60:57} $$

$$ \bar{A}_{60} = \int_0^\infty {}_t p_{60}\, \mu_{60+t}\, e^{-\delta t}\, dt $$

$$ = 0.04 \int_0^\infty e^{-(0.04+0.05)t}\, dt $$

$$ = 0.04 \left[ \frac{e^{-0.09t}}{-0.09} \right]_0^\infty $$

$$ = \frac{0.04}{0.09} (1-0) $$

$$ = 4/9 $$

Similarly $\bar{A}_{57} = \dfrac{0.03}{0.03+0.05} = 3/8$

and $\bar{A}_{60:57} = \dfrac{0.07}{0.07+0.05} = \dfrac{7}{12}$

Q5

Then $\bar{A}_{\overline{60:57}} = \frac{4}{9} + \frac{3}{8} - \frac{7}{12}$

$= \frac{17}{72} = 0.23611$

(b) $E(e^{-\delta \bar{T}_x})^2 = E(e^{-2\delta \bar{T}_x})$

So repeat (a) with $2\delta = 0.10$ instead of $\delta$.

$^2\bar{A}_{60} = \frac{0.04}{0.14} = \frac{2}{7}$

$^2\bar{A}_{57} = \frac{0.03}{0.13} = \frac{3}{13}$

$^2\bar{A}_{60:57} = \frac{0.07}{0.17} = \frac{7}{17}$

$\therefore {}^2\bar{A}_{\overline{60:57}} = \frac{2}{7} + \frac{3}{13} - \frac{7}{17}$

$= 0.10472$

$\therefore$ Variance $= 0.10472 - \left(\frac{17}{72}\right)^2$

$= 0.04897$

SD $= \sqrt{\quad}$ //

$= 0.22129$

Q6

(a) ${}_{t-1}p_{65} = 0.99^t$

Profit signature = Profit vector $\times\ 0.99^t$

$= (200,\ -118.80,\ -9.80,\ 29.11)$ ①

Reserve required $= 118.80\,v^1 + 9.80\,v^2$ at 7%

$= £119.59$

Then $200 - 119.59 = 80.41$,

so profit signature $= (80.41,\ 0,\ 0,\ 29.11)$

(b) ${}_{t-1}p_{65} = 0.9 \times 0.99^t$ for $t \geq 2$

Profit signature as ①, but $\times 0.9$ for $t \geq 2$

$\therefore$

Profit signature for $t = 1$; extra $70 \times 0.1$ for profit on surrender, giving $200 + 0.1 \times 70$

$\therefore$ ① becomes $(207.00,\ -106.92,\ -8.82,\ 26.20)$

Reserve required $= 107.63$

so profit signature $= (99.37,\ 0,\ 0,\ 26.20)$

(c) Discounting at 12% gives $(71.80 + 0 + 0 + 18.50 = 90.30)$

and $(88.72 + 0 + 0 + 16.65 = 105.37)$

Q6

(d) Difference = 15.07

(i) Profit on surrenders in year 1 (= 7)

(ii) For surrenders, the losses in years 2 and 3 disappear

( > profit in year 4)

Q7

(a)

$$PSL = 500 \times \frac{8}{90} \times 45000 \times \frac{{}^{128026}\bar{M}^{r_a}_{43}}{S_{42.5} D_{43}} = \$11,904,826$$

$S_{42.5} = 8.1905$, $D_{43} = 2626$

(b)

$$FSL = 500 \times \frac{1}{90} \times 45000 \times \frac{{}^{2500182}\bar{R}^{r_a}_{43}}{S_{42.5} D_{43}} = \$29,060,728$$

(c) $TSL = PSL + FSL$

$= \$40,965,554$

$$\text{Contribution of } 1\% = 500 \times \frac{45000}{100} \times \frac{{}^{294196}\bar{N}_{43}}{S_{42.5} D_{43}} = \$3,077,614$$

$$\therefore k = \frac{40,965,554 - 9,000,000}{3,077,614}$$

$$= \underline{\underline{10.39\%}}$$

## Comments for external examiner

Q1 is bookwork.

Q2 is similar to a tutorial question.

Q3 is bookwork.

Q4 is unseen. The formula for the (b) was mentioned in lectures.

Q5(a) is bookwork. (b) is unseen.

Q6 (a) is standard. (b) is unseen. (c) is standard.

Q7 is similar to a tutorial question.